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Dynamic photovoltaic building envelopes for adaptive energy and comfort management

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Supplementary Information

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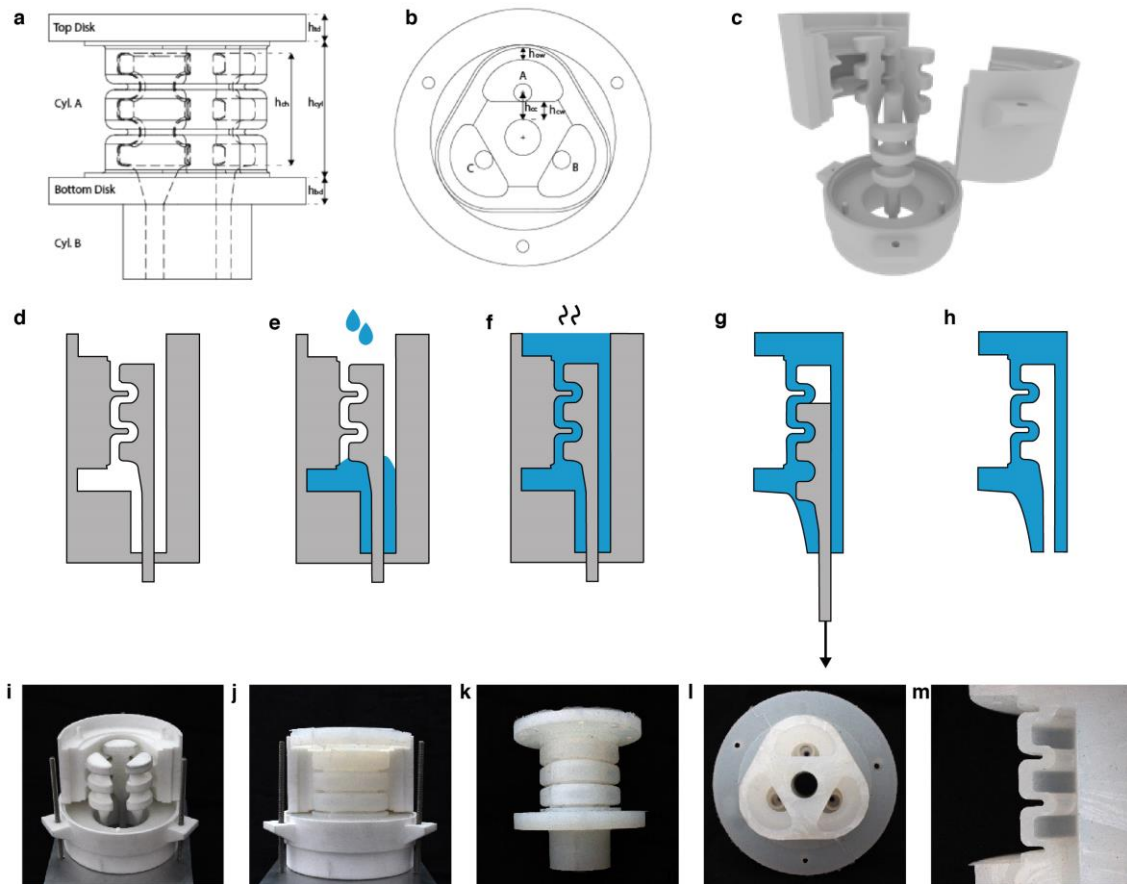
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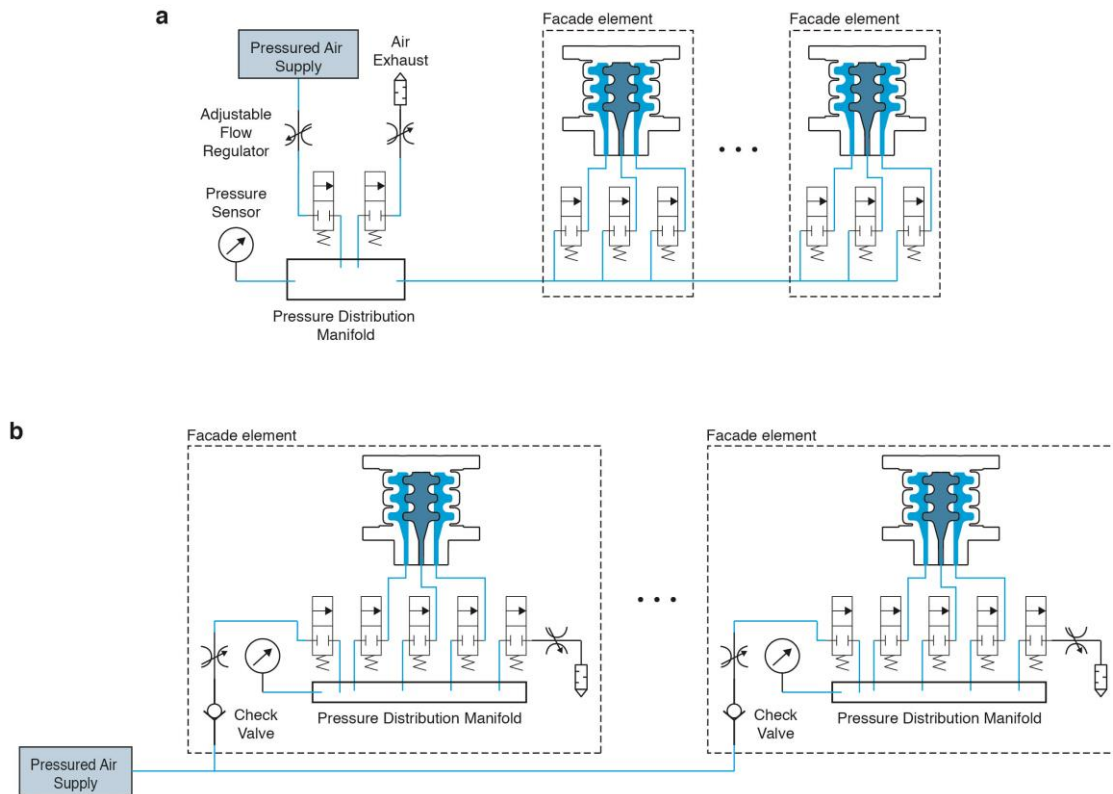
Supplementary figures



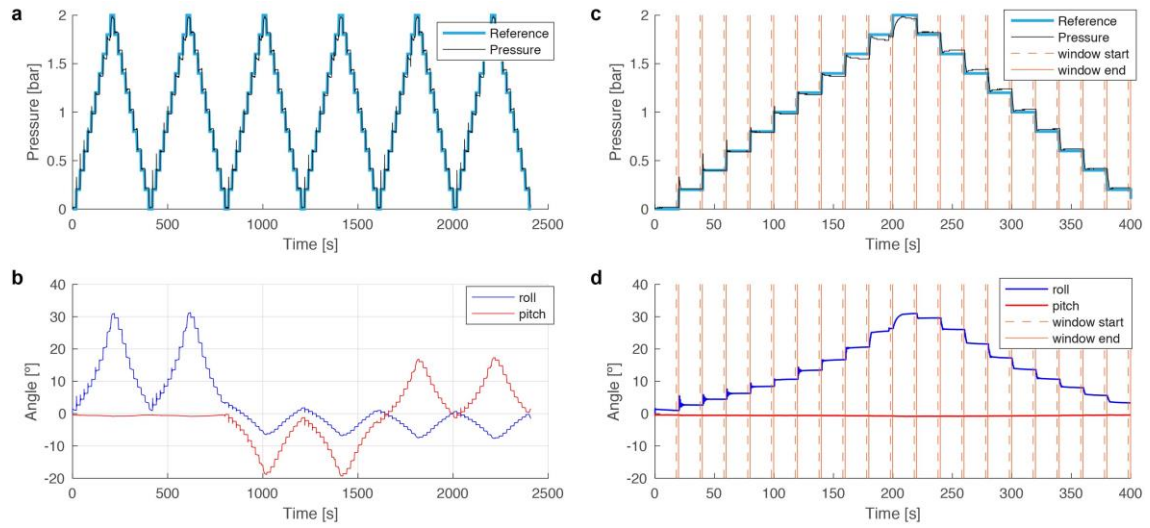
Supplementary Figure 1 | A quick-cast methodology for fabrication of soft

actuators. **a**, Vertical cross section of the two-axis three-chambers soft actuator with the ribbed outer shape and internal cavities. **b**, Horizontal cross section of the soft actuator. **c**, 3D model of the mould. **d**, Schematic representation of a mould with one

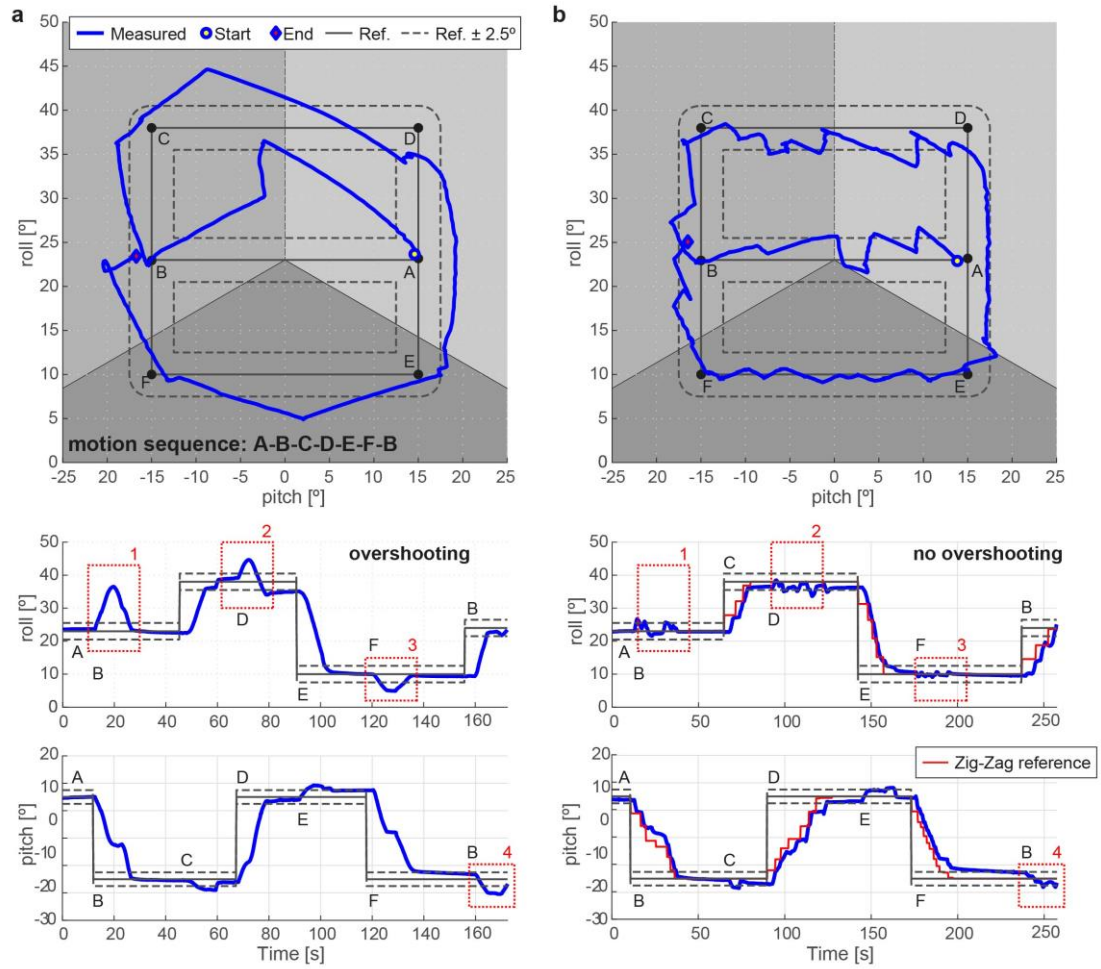
bullet-chamber. **e**, Pouring rubber in liquid state. **f**, Degassing and curing. **g**, Pulling out bullet-chamber to form a cavity in the rubber. **h**, Single-body soft actuator with complex and precise ($< 1\text{mm}$) geometry of the outer shape and cavity. **i**, Image of the Nylon mould assembled on the aluminium plate. The front part of the mould is missing in the image. **j**, Soft actuator with the mould after disassembling the front part of the mould. **k**, Final unitary body soft actuator made from Elastosil® Vario fabricated in a single casting step. **l,m**, Images of the horizontal and vertical cross sections of the soft actuator, respectively, showing the achieved complex wall shape.



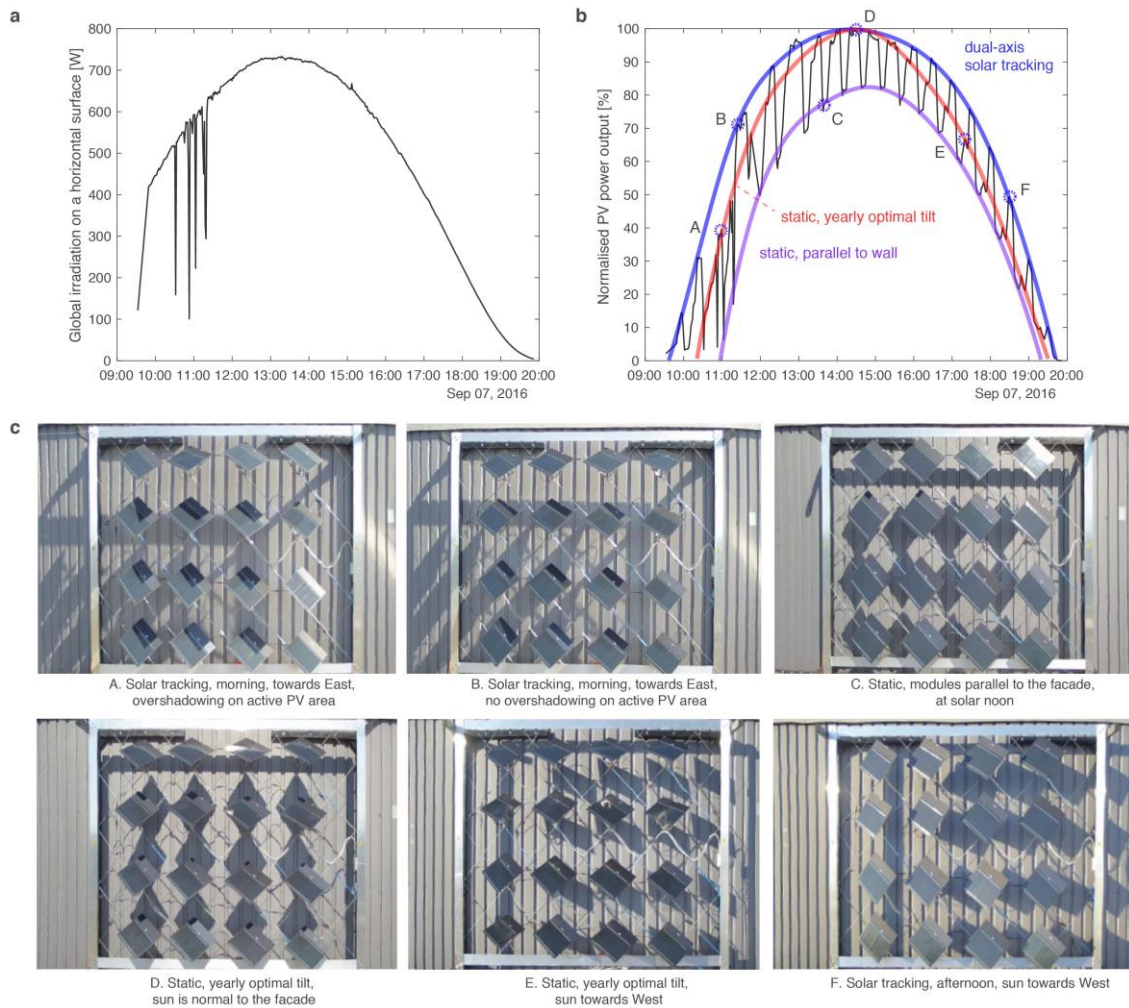
Supplementary Figure 2 | Different complexities of decentralised, modular pneumatic control system. a, Decentralised pneumatic control system of minimal complexity per envelope element, with a single common pressure regulation chamber for all modules. This configuration offers limited freedom in parallel control of the soft actuators. **b,** Decentralised pneumatic control system of maximal complexity per envelope element. Each envelope element is equipped with a pressure regulation chamber. Therefore, envelope elements can be independently controlled.



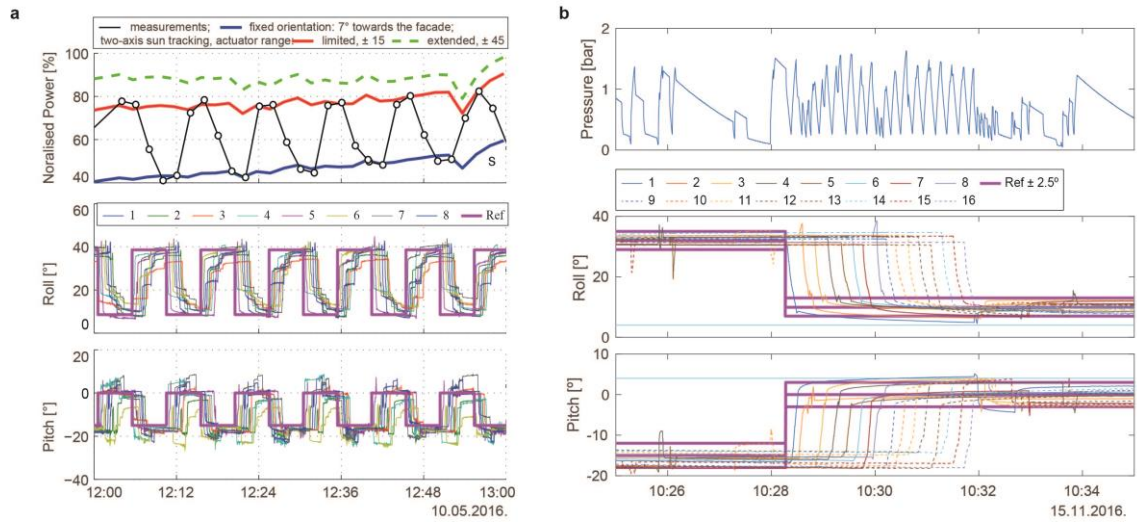
Supplementary Figure 3 | Quasi-static pressure-deflection measurements and extraction of characteristic steady-state points. Pressure increasing-decreasing cycles (a) and corresponding angles (b) repeated two times for each of the three chambers of the soft actuator. Extraction of steady-state points from a single pressure increasing-decreasing cycle (c) and corresponding angles (d). Data was taken as an average of the last 2s in the 20s step.



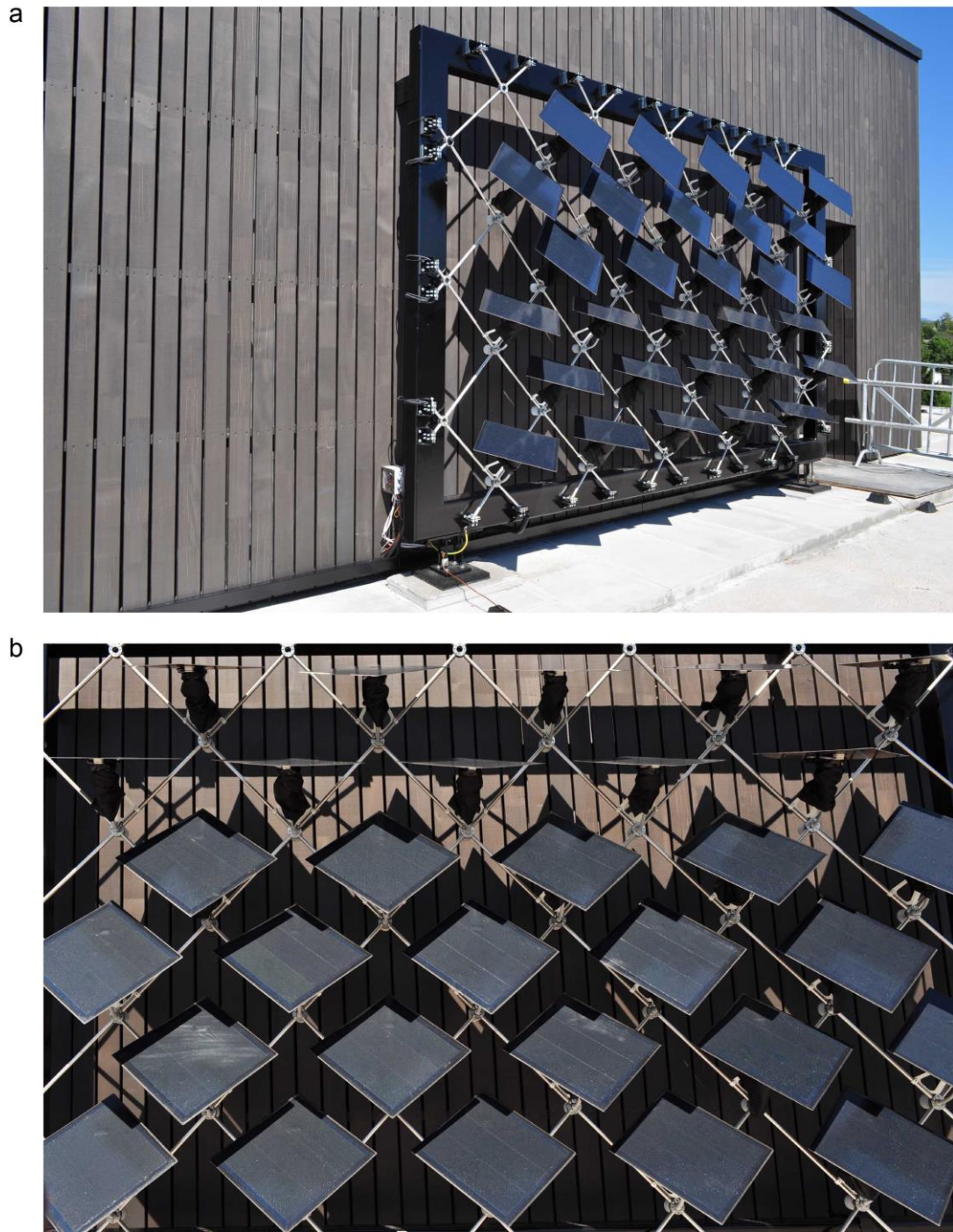
Supplementary Figure 4 | Orientation feedback control of the soft actuator performing the figure-of-eight reference tracking. a, Roll-pitch and time diagrams of the zig-zag reference tracking without an upper bound on the maximum distance from the reference orientation. **b**, Roll-pitch and time diagrams of the zig-zag reference tracking with an upper bound on the maximum distance from the reference orientation. Introducing the upper bound on the maximum distance from the reference orientation reduces overshooting, as indicated by red dotted boxes.



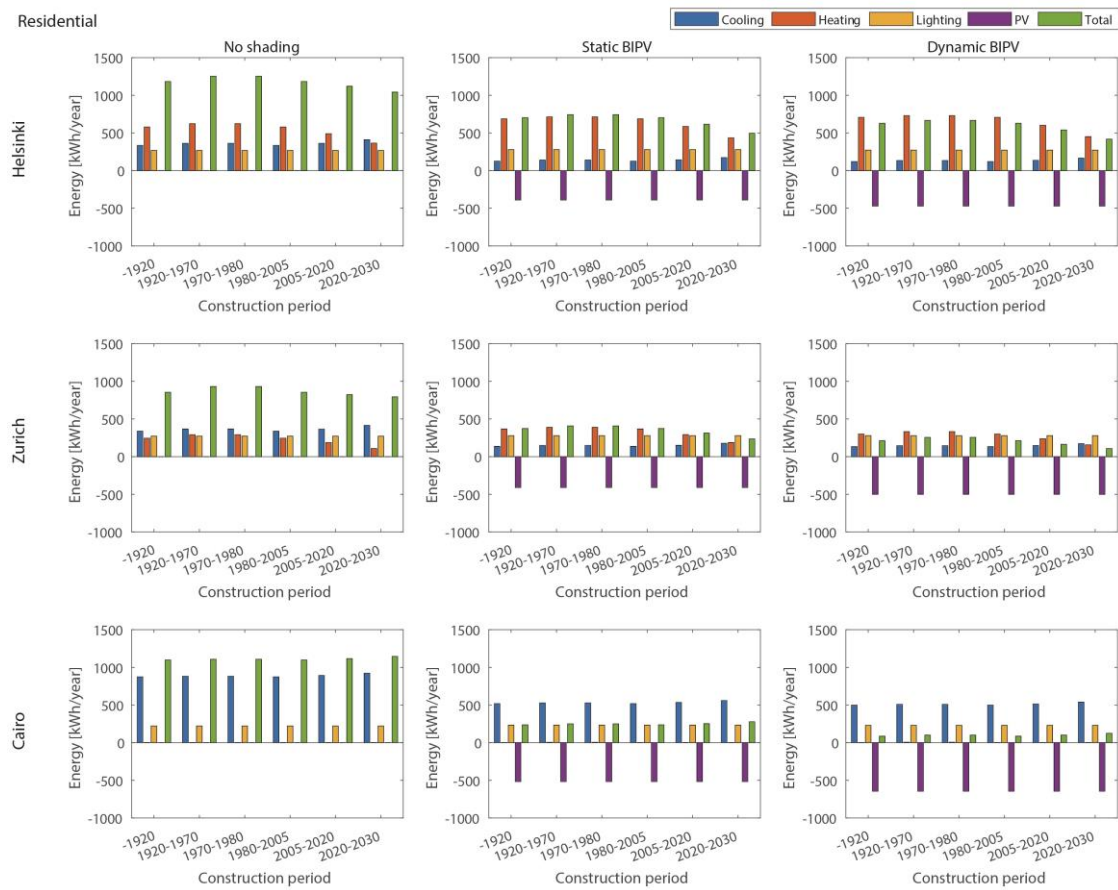
Supplementary Figure 5 | Solar tracking experiment with the dynamic envelope with 16 modules. The envelope is mounted on the roof of HPZ building at ETH Zurich, Switzerland. **a**, Measurements of the global solar irradiation on a horizontal surface. **b**, Comparison of PV power output of the envelope in three states: two-axis solar tracking, static facade parallel to the wall, and static facade tilted at 45°. The envelope cycles through the three states every 30 min (10 min per state). **c**, Characteristic moments of envelope orientation during the solar tracking experiment.



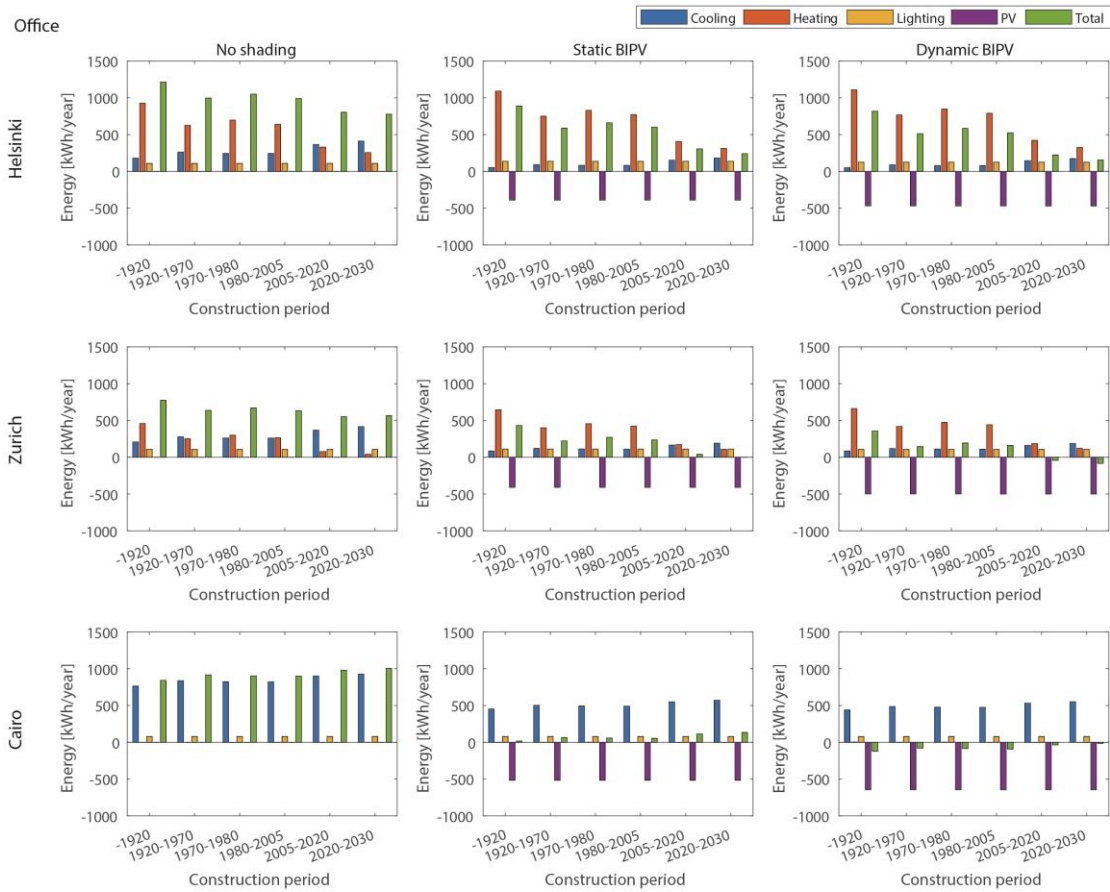
Supplementary Figure 6 | Sequential control of envelope elements. a, Sequential control of the envelope with 8 modules cycling between two states: parallel to the wall and solar tracking. **b,** A single transition step of the sequential control of the envelope with 16 modules. Both experiments were performed using the HPZ envelope.



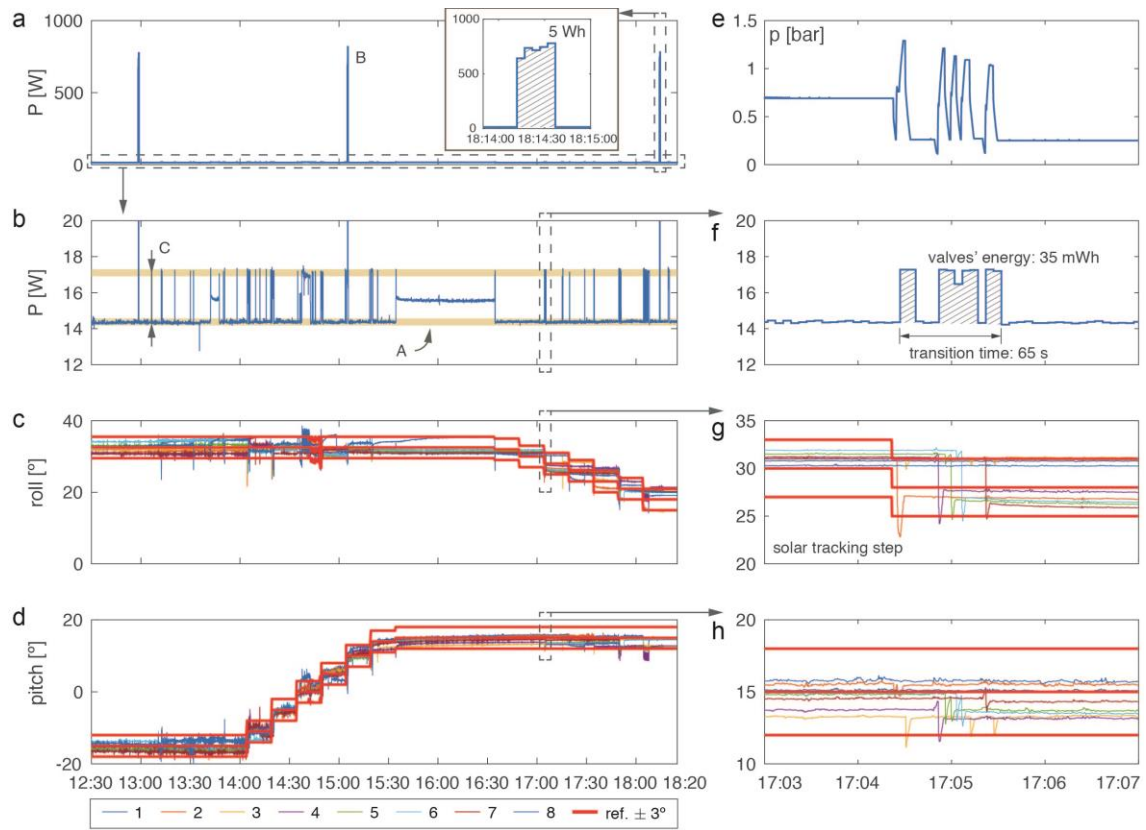
Supplementary Figure 7 | Soft robotic driven dynamic envelope at HiLo. a, Perspective view on the dynamic solar envelope. The top three rows are closed, while the bottom three rows are open. **b,** Front view on the dynamic solar envelope with the top rows open.



Supplementary Fig. 8. Heating, cooling, lighting demands and PV generation for three envelope cases, in three cities, and over six construction periods for the residential case.



Supplementary Fig. 9. Heating, cooling, lighting demands and PV generation for three envelope cases, in three cities, and over six construction periods for the office case.



Supplementary Figure 10 | Electricity consumption measurements of the pneumatic control system during solar tracking experiment. For this experiment, lower half (8 modules) of the HPZ envelope was used. **a**, Aggregated power consumption of the pneumatic control system consisting of the control electronics (A), air compressor (B), and electro-pneumatic valves (C). **b**, Zoomed in base power consumption from the electronics and variable power consumption from the valves. **c,d**, Roll and pitch angles during the solar tracking experiment. **e-h**, Zoomed in pressure in the pressure distribution chamber (**e**), aggregated power consumption (**f**), roll (**g**) and pitch (**h**) angles during one solar tracking step.

Supplementary Note 1

The orientation control of the soft actuator is based on the measurements of the roll and pitch angles using the IMU sensors InvenSense™ 9-DOF MPU9250, mounted on a breakout board from Drotek Electronics (www.drotek.fr). The IMU sensors are connected via I2C bus to a small single-board computer Raspberry Pi 2 Model B. The IMU sensor data is transferred to the Raspberry Pi and a fusion algorithm¹ is ran at 40 Hz, updating the orientation angles. The orientation angles are sent as user datagram protocol (UDP) packets over the local area network (LAN) to NI PLC. Orientation angles are used in the real-time control loop programmed in LabVIEW Real-Time, running at 20 Hz.

The Zig-Zag control algorithm has been developed in order to map the three available actuation directions (Fig. 2f), generated by the individual inflation of the three chambers, to two angles, pitch and roll². Starting from the initial position in roll-pitch plane, the algorithm computes which chambers and in which way they need to be inflated in order to reach the reference orientation. If the current orientation lies on the main direction of actuation of a certain chamber, only that chamber will be used to move the panel. If the current orientation lies in between two main directions of actuation, both that chambers will be used to rotate the module. Due to the simplicity of the pneumatic control system, where only a single pressure regulator exists for a single actuator, the two chambers are used sequentially, one at a time. This generates zig-zag motion of the facade module, shown in Supplementary Figure 4a. If the soft actuator was supposed to follow the reference trajectory consisting of straight lines between some points, for example, as in the figure-of-eight (A-B-C-D-E-F-B) reference tracking, large overshoots and drifting away from the straight trajectory occur. This overshoot could be reduced by introducing an upper limit on the maximum difference between the pitch and roll angles and the straight line connecting the initial and final orientation. When such a limit is introduced, the overshoot can be drastically reduced. Experimental results show that the control algorithm was able to respect the upper bound of $\pm 2.5^\circ$ at almost all times during the transition phase (Supplementary Figure 4b). The small overshoots of about 1° occur for a very short time due to the bang-bang control strategy used. This could be avoided if more sophisticated control algorithms were used, such as proportional integral derivative (PID) control or model predictive control³. Due to the narrower angle band, the switching between two chambers occurs more often, which

extends the transition time from 160s to 250s for following the whole trajectory. As the solar tracking requires updating the orientation every 5-10 min for only few degrees, the time difference introduced by setting the upper limit on the tracking precision is negligible. The algorithm is available in an open access GitHub repository⁴.

Supplementary Note 2

In this comparison, the static BIPV façade uses the same technology as the dynamic system, with the panels fixed at an optimal angle. Buildings in the Helsinki climate have a large heating demand and only little cooling demand. In this case, the application of either a static or adaptive BIPV façade system only has a small impact on the overall performance of the building. Buildings in Cairo, which are dominated by cooling demand, show very large savings for both, a static and a dynamic BIPV façade system, with slightly higher savings in case of the dynamic facade. As the building is almost always in a cooling mode, the panels in the dynamic case will always move to a position that is very similar to the already optimised static configuration. Zurich, on the other hand, has large variations in climate both seasonally and during the day. The dynamic system is therefore able to adapt to these changes and show savings both in comparison to the base case without a shading system, and to a statically optimised system.

For the simulation, the lighting load of office and residential buildings are set to 15.9 W/m^2 and 10.5 W/m^2 using a control threshold of 300 lux and 250 lux, respectively. Due to the large window opening with respect to the floor area, the assumption of no external buildings obstructing the path of sunlight, and the south facing orientation, natural lighting is capable of providing most of the lighting requirements. Furthermore, the occupancy schedule of the office building follows standard working hours that are often during sunlit conditions. Both factors result in a rather low lighting energy demand of the room. Variations of these parameters can have large impact on the overall performance, which we have previously discussed in more detail⁵.

Note that according to the archetype database⁶ used (the data is available in a public repository, the link is provided in the references), buildings constructed in the 1970s-1980s exhibit a higher overall envelope U-value than buildings built beforehand. This explains the slight increase in heating demand during this period.

Supplementary Note 3

The electricity consumption of the pneumatic control system of the experimental dynamic envelope at HPZ building (Supplementary Figure 10) was measured during a solar tracking experiment using smart AC meter SINEAX AM1000 from Camillie Bauer AG. Note that during this experiment the system was just tracking the sun, without going to the other states of the envelope, as defined in the Methods section on Solar tracking. Also, notice that only the lower half of the HPZ envelope (8 modules) was used in this experiment. The orientation update cycle was 15 min. We measured the aggregated AC power consumption of all the components forming the pneumatic control system, which are the air-compressor Vento Silent OM200 from ABAC with the rated power of 750 W, 24V DC 2-2/way valves from FESTO with rated power of 1 W, NI cRIO 9074 PLC with the stand-by power consumption of 12 W, and Raspberry Pi B+ with the power consumption of 2 W, to which IMU sensors were connected.

There are three characteristic components of the AC consumption graph related to the three main components of the pneumatic control system (Supplementary Figure 10a). There is a constant, base, low power consumption, coming from the control electronics and pneumatic valves, and short peaks of high power consumption from the air-compressor. There is a constant power consumption of the control electronics, NI PLC and Raspberry Pi, of about 14.3 W. The variable low power consumption from the pneumatic valves adds to the power consumption of the electronics as a variable part with the maximum at 17.5 W. The air compressor generates peaks of about 750 W for about 45s, when it refills the tank from 6 to 8 bar. One solar tracking transition step is shown in zoomed-in graphs (Supplementary Figure 10e-h). The transition step lasted 65s and valves used 35 mWh. These values are used to estimate the yearly energy consumption of the experimental control setup (Supplementary Table 1).

As indicated in the table, the majority of the total power consumption (34.4W) of the pneumatic control system is used by the control electronics (22.5W, or 65%). This is due to the fact that the control electronics was not selected with intention of low power consumption and final implementation of the control system, but rather for quick development of the control system and control software implementation. Therefore, there is a huge potential to design much more efficient pneumatic control system with off-the-shelf components with intention for a final product design. A control electronics

board could be developed utilising ultra-low-power microcontroller, such as one from Texas Instruments MSP430™ family. For example, MSP430FR59941 requires 0.4 mW in normal regime, 0.1 mW in stand-by with very fast wakeup time of 7µs, and it has enough computational power and number of inputs and outputs to control the envelope. Also, the envelope control algorithms are not computationally intensive and run at 5 Hz. This means that the power consumption of the control electronics could be reduced to around 1% of the current power consumption; however, due to additional Ethernet communication electronics for monitoring of the envelope, we assumed that the power consumption of the control system is reduced to 10% of the values for experimental setup at HPZ roof.

Further, it was found that only 60% of the power consumed by the ABAC air-compressor was used in the pneumatic system; other 40% was wasted due to the leakage in the compressor. This stand-by leakage of the compressor could be avoided to a large part by using an air-compressor with the sealing of a higher quality. Further, more energetically efficient electro-pneumatic valves are available on the market from SMC Pneumatics S070 series with only 0.1 W power consumption, instead of the current 1 W of Festo MHA1 series. In order to obtain the final energy balance of the envelope, we compare the electrical energy that the envelope can generate over the year with the electrical energy it uses for its operation. The generated electrical energy is calculated based on the solar irradiation data for south oriented façade surface in Zurich⁷. Based on the above measurements, the self-consumption of the experimental control setup was estimated at 30%. If the control setup is designed for low-power consumption using only off-the-shelf components as explained above, the energy consumption of the pneumatic control system is estimated at 3% or less (see Supplementary Table 1). This number further depends on the climate region; in climates with more solar irradiation, the self-consumption of the envelope will be lower than 3%, due to higher solar irradiation per m². Also, if thin-film PV cells with the efficiency higher than 10% are used, this will further reduce the self-power consumption below 3%.

Supplementary Table 1 | Energy balance of the soft-robotics-driven dynamic building envelope: electricity generation vs. self-consumption. Calculations are for the HiLo envelope with 30 elements. Regarding the energy consumption, it is assumed that the envelope is active only during day-lit hours; there are 4,080 day-lit hours in Zurich in a year.

Electricity generation			
Solar irradiation on a South facing façade in Zurich, Switzerland [kWh/m ² /year] [§]	960		
Number of envelope modules	30		
PV module area [m ²]	0.16		
Efficiency of thin-film PVs [%]	10		
(A) Electricity generation [kWh/year]	460.8		
Self-consumption			
	Average power consumption [W]	Energy consumption [kWh/year]	Percentage of the yearly production [%] [‡]
Experimental setup[†]			
Control electronics (NI PLC, Raspberry Pi)	22.5	91.7	20
Air-compressor, in constant solar tracking	8.4	34.4	7
Leakage	1.5	6.1	1
Electro-pneumatic valves (Festo MHA1 series)	2.0	8.1	2
Total energy consumption:	34.4	140.3	30
Control system optimized for energy consumption (using only off-the-shelf components)			
Control electronics (ultra-low power microcontrollers)	2	8.2	2
Air-compressor, in adaptive control [◊]	/	3.5	1
Leakage (assumed 10% of the air used) [¶]	/	0.4	<1
Electro-pneumatic valves (SMC S070 series)	0.2	0.8	<1
Total energy consumption:	/	12.83	3

[§] Geographical Assessment of Solar Resource and Performance of Photovoltaic Technology⁷

[†] Experimental setup at HPZ roof. The consumption values are measured (see Supplementary Figure 10) and then scaled up to represent values for an envelope with 30 modules.

[‡] Calculations with respect to (A) in the Table.

[◊] Performing minimisation of the building net energy consumption (see Section on *Energy benefits of the envelope as part of the building service system.*)

[¶] Page 9, Compressed Air Evaluation Protocol, National Renewable Energy Laboratory⁸

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